

BUSINESS PROPOSAL / TEAM YBQZYHP

StockPilot

Purchasing intelligence, explained.

NUS-ISS Show Me Your Agent Hackathon
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01 / THE BUSINESS CASE

The problem

Small retail operators need to decide what to reorder, how much to buy and which supplier can deliver in time. When sales history, inventory and supplier terms are reviewed separately, purchasing staff must reconcile them manually. A low unit price can hide a late delivery, while a large precautionary order can tie up cash. Managers need a clear basis for approving the final purchase.

Our proposal

StockPilot brings these decisions into one explainable purchasing workspace. It uses sales and inventory CSVs to forecast demand, flag projected shortages, compare supplier allocations and prepare purchase-order drafts for human review. The initial use case is a small catalogue of 10-15 products, demonstrated with 12 products and three simulated approved suppliers.

Value to the business

The intended benefits are faster preparation, earlier visibility of stock gaps and better cost-versus-delivery decisions. Every recommendation should answer four questions: why this quantity, why these suppliers, what could break the plan and who reviewed it. These benefits are hypotheses for a pilot; commercial savings and real-store outcomes have not been established.

Primary users	Proposed buyer
Purchasing staff and owner-operators preparing replenishment decisions.	The owner or operations manager of a small retail business.

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02 / SOLUTION & DIFFERENTIATION

From stock data to a reviewable order

StockPilot connects the purchasing workflow while keeping calculations inspectable and the approval decision with a person.

Workflow	What StockPilot does	Business purpose
1. Prepare data	Validate sales and inventory CSVs; retain a clearly labelled sample option.	Start from consistent inputs and surface data gaps.
2. Anticipate demand	Forecast 14 days using recent weekday patterns; show stock coverage and owner-entered adjustments.	Identify products needing intervention before a purchase is drafted.
3. Compare sourcing	Evaluate one- or two-supplier allocations against delivery timing, price, quantity and eligibility constraints.	Explain the trade-off between early availability and purchasing cost.
4. Check resilience	Replay the proposed allocation under higher demand or delayed new deliveries.	Expose assumptions that could make the plan fail.
5. Review and record	Apply a shared-budget draft gate, persist order details and support simulated manager review and exports.	Provide a reviewable decision and a record of the approved simulation.

A bounded agent with evidence

The optional AI briefing follows a controlled sequence: read inventory, read recommendations, check budget, then explain the results. Python performs forecasting, purchasing arithmetic and constraint checks independently. The model has no approval or dispatch tool. Live organiser-gateway connectivity remains unverified.

Why this approach is useful

The proposition combines demand, delivery timing, sourcing and review in one small workflow. A recommendation includes its assumptions and blocked conditions, rather than a quantity alone. Compared with the proposed manual-workflow baseline, the pilot will test whether this improves preparation time and decision quality; no claim of competitor superiority is made.

Current boundary: Supplier data and manager roles are simulated. The prototype does not send orders or emails, reserve inventory or confirm incoming stock after approval.

03 / ADOPTION & COMMERCIAL MODEL

Start with a focused retail pilot

Initial customer profile

Recruit small retailers with repeat purchasing, accessible daily sales exports and a manager able to review recommendations. Begin with 10-15 regularly traded products, where stock records and supplier lead times can be checked. This is a proposed beachhead, not a validated market-size estimate or a claim of customer demand.

Route to adoption

Use a short workflow demonstration to recruit a small pilot cohort through direct retailer outreach and relevant business networks. Collect the current purchasing process, then run StockPilot alongside it. Start in shadow mode: staff compare the recommendations with their usual decisions while normal purchasing controls remain in place. No pilot partners or distribution agreements are currently claimed.

Commercial hypothesis	How to validate it
A subscription per store or small catalogue, with optional paid onboarding.	Interview owners about willingness to pay after demonstrating a useful, repeatable workflow. Set pricing only after validating value and support effort.
CSV-based onboarding reduces the initial integration burden.	Record time spent mapping exports, correcting inventory and maintaining supplier data.
Explainability and human review support repeat use.	Track weekly usage, overrides, rejected drafts and the reasons managers accept or reject recommendations.

Illustrative customer economics

For planning only, assume 20 purchasing cycles per month, 15 minutes saved per cycle and staff time valued at S\$20 per hour. The implied time value is $20 \times 15 / 60 \times \text{S\$}20 = \text{S\$}100$ per month. This is an assumption-based illustration, not a measured saving or necessarily a cash reduction. Purchasing savings should be counted separately only when service outcomes remain comparable.

Delivery cost and sustainability

The main cost drivers are hosting, model requests, onboarding, data cleanup and ongoing support. The core calculation workflow runs without an API key; AI briefings are optional and bounded to a short sequence. During the pilot, record actual usage and support effort before setting a price or projecting margins. AWS hosting is a proposed delivery option; deployment evidence is a separate submission item.

04 / SUCCESS MEASURES

Test value before claiming impact

The numerical targets below are proposed acceptance goals inherited from the project scope. They are not achieved outcomes. Baselines, sample size and observation periods should be agreed before the pilot begins.

Measure / proposed target	Measurement approach
Preparation time Under 5 minutes per agreed batch	Time a person from validated inputs to a review-ready draft. Record data cleanup separately and compare with the existing workflow on similar batches.
Forecast error Aggregate WAPE below 20%	Use held-out sales dates with training restricted to earlier dates. Compare weekday averages, last-week forecasts and moving averages; also report per-product MAE and nonzero-actual MAPE.
Early warning 7 days where observable	Record when a projected shortage is first flagged and the eventual shortage date. Report missed and false alerts. Seven days is a target, not a guarantee when inputs arrive late.
Purchasing cost 5-10% reduction hypothesis	Compare cost with the cheapest feasible single-supplier plan under the same constraints. Report service outcomes alongside cost; do not claim savings when no comparable feasible plan exists.
Review and adoption Track throughout the pilot	Record draft completion, manager decisions, overrides, stockout days and unmet units. Use these to detect faster but poorer decisions.

What the prototype can evaluate now

The offline evaluator uses a final 14-day holdout and at least 42 consecutive daily observations. It compares three forecast methods and replays purchasing against a reorder-point baseline. The bundled data contains 84 synthetic daily observations for each of 12 products. This supports reproducible demonstrations, not claims about current Singapore retail demand.

Fair interpretation

The reorder-point replay and StockPilot may deliver different service levels, so lower simulated cost alone does not establish a benefit. Calculation runtime is also different from human preparation time. No numerical evaluation result or real-store improvement is claimed in this proposal.

05 / DELIVERY PLAN & READINESS

A clear path from demo to pilot

Proposed stage	Exit condition
1. Harden the prototype	Resolve the known calculation findings, validate urgent-order comparisons and rehearse the complete review workflow.
2. Establish a baseline	Agree the pilot catalogue, verify inventory and supplier assumptions, document the current process and define success measures.
3. Run in shadow mode	Compare recommendations with ordinary decisions, record manager feedback and evaluate time, cost and service together.
4. Decide on a controlled pilot	Proceed only after correcting material issues, validating value and adding authenticated access, reliable operations and order reconciliation.

Current delivery evidence

The repository documents CSV validation, catalogue analysis, forecasting, supplier splitting, budget checks, disruption scenarios, persisted drafts, simulated review and exports. Offline evaluation and mocked gateway integration are included. This is a hackathon prototype, not a production-readiness certification.

Material limitations to resolve

Three calculation findings remain open: percentage rounding may overstate demand; supplier comparison may omit an urgent deadline; and one zero-quantity conflict message may misidentify incoming stock as the cause. Real supplier behaviour, authenticated approvals, overlapping-order reconciliation and catalogue scaling require further work.

Hosting and pilot readiness

Live gateway connectivity and AWS deployment are unverified in the reviewed documentation. A hosted pilot needs access controls, HTTPS, backups, monitoring and an operating owner. Supplier dispatch is outside the prototype scope.

The proposed next step

Seek retailer pilot participation and feedback to establish measurable value before investing in deeper integrations or commercial rollout.

PROJECT BASIS & REFERENCES

Repository: github.com/xensica/nus-iss

Project and purchase_demo READMEs: features, evaluation and limitations. Purchasing-agent requirements: intended scope; older status labels are superseded by the current application guide.

Commercial model, economics and rollout are planning assumptions, not achieved results.